Statistical Analysis and Predictive Feature Engineering of California Seismicity (1995–2025)

Course Project for EP4130 Data Science Analysis

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Abstract

This study presents a comprehensive statistical and machine learning framework for analyzing seismic activity and identifying predictive precursors to major earthquakes using earthquake catalogs from the region of California (1995–2025). The primary objective is to develop physically interpretable and statistically robust features that capture underlying tectonic stress dynamics. After applying data quality filtering and declustering techniques to ensure independence of events, baseline seismicity is characterized using the Gutenberg–Richter law and Poisson statistics. Maximum Likelihood Estimation (MLE) is used to yield an accurate regional b-value (Gutenberg Richter Law) of approximately 0.81.

Temporal analysis of the rolling b-value reveals distinct stress accumulation phases preceding the 2019 M7.1 Ridgecrest earthquake, highlighting a rapid short-term "seismic ignition" mechanism rather than a gradual long-term precursor. Building on these insights, multiple spatiotemporal features—including rolling b-value, event rate, spatial variance, and cumulative Benioff strain—are engineered and integrated into an XGBoost-based predictive framework.

The model achieves a recall of 0.73 and an ROC-AUC of 0.862, demonstrating strong capability in detecting moderate-to-large earthquake events ($M \geq 4.0$), although with low precision due to inherent class imbalance. These results validate the predictive value of physics-informed features and highlight the trade-off between sensitivity and false alarms in seismic forecasting.

Overall, this work bridges statistical seismology and machine learning, demonstrating that physically grounded feature engineering can significantly enhance earthquake prediction frameworks and provide actionable insights for seismic hazard assessment.

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1 Introduction

The lithosphere is composed of huge tectonic plates, which forms the outermost layer of the Earth. These tectonic plates *float* on top of the asthenosphere, which is composed of lava. Occasionally, the tectonic plates collide with each other causing an outburst of energy into the lithosphere. This manifests as intense shaking of the ground called as an Earthquake. Earthquake prediction is a challenging problem because the Earth is a complex and highly dynamic system. However, seismic activity (collective term for earthquake shocks) does not occur randomly in isolation; earthquakes tend to appear in patterns over both space and time. Understanding these patterns is essential for identifying signals that may indicate an upcoming major event.

1.1 Types of Seismic Events

Earthquakes typically occur as part of a sequence rather than as single, independent events. These sequences can be classified into several types:

- **Mainshock:** The largest earthquake in a sequence, representing a major release of built-up stress along a fault.
- **Foreshocks:** Smaller earthquakes that occur before the mainshock in the same region. These may indicate the early stages of fault failure.
- **Aftershocks:** Events that follow the mainshock as the surrounding crust adjusts to the new stress distribution.
- **Seismic Swarms:** A group of earthquakes without a single dominant event, often linked to fluid movement rather than tectonic stress.
- **Seismic Quiescence:** temporary decrease in small earthquakes, which may suggest that stress is building up in a locked fault.

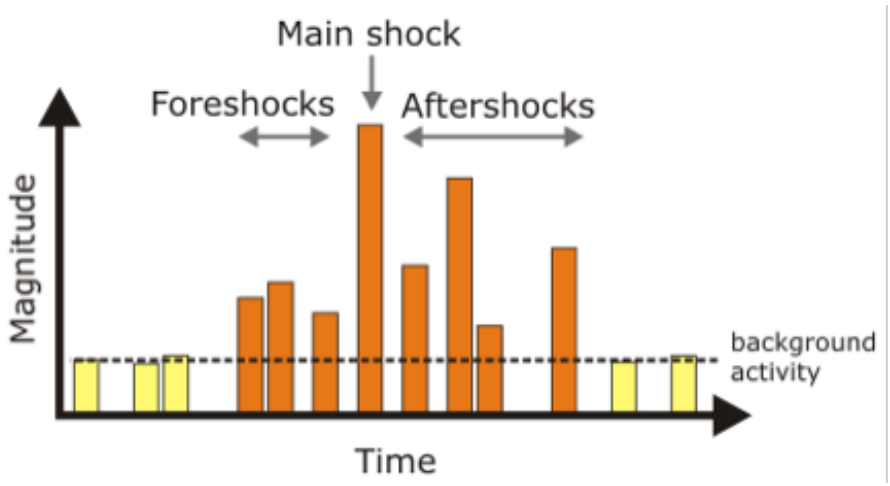

Figure 1: A demonstrative graph showing foreshocks, the mainshock and the aftershocks

1.2 The Gutenberg-Richter Law

The most fundamental principle of statistical seismology is the Gutenberg-Richter (GR) law. It is an empirical power-law distribution that relates the magnitude of an earthquake to its

frequency of occurrence. The relationship is defined as:

$$\log_{10} N(M) = a - bM \tag{1}$$

Where:

- $N(M)$ is the **cumulative** number of earthquakes with a magnitude greater than or equal to M.
- a (**Seismic Productivity**): Represents the overall background activity level of the region. A higher a-value means a higher absolute number of earthquakes, regardless of size.
- b (**The b-value**): The slope of the distribution, dictating the ratio of small to large events. A normal baseline value for active tectonic crust is $b \approx 1.0$, meaning for every M 4.0 event, there are approximately ten M 3.0 events and one hundred M 2.0 events.

Predictive Significance of b: The b-value acts as an inverse stress meter. If b drops significantly ($b < 1.0$), the frequency distribution flattens. This indicates a "deficit" of small earthquakes relative to large ones, implying the fault is locked and accumulating high stress—a classic warning sign of an impending major rupture. Conversely, if $b > 1.0$, the slope is steep, indicating the earth is easily releasing energy through many tiny fractures (often seen in volcanic swarms or creeping faults).

Now, the plan of action is as follows:

1. Collect Earthquake data
2. Perform a quality check and data sanity
3. Fit the Gutenberg Richter Law
4. Explore the spatial dynamics
5. Predictive modeling
6. Model Evaluation

2 Data Quality and Uncertainty Analysis

2.1 Data Sources and Temporal Coverage

The seismic data utilized in this study were retrieved from the **USGS Earthquake Catalogue**.

- **California Catalog (1995–2025):** A 30-year high-resolution dataset focusing on the transform boundary system. This catalog serves as the primary dataset for our baseline analysis and predictive feature engineering due to its consistent station density and well-documented fault geometries.

Datasets were downloaded as csv file with key columns being: latitude, longitude,depth, mag,magType, type, horizontalError, depthError and magError.

2.2 Selection of the California Dataset for Primary Analysis

1. **Tectonic Consistency:** California is mainly characterized by strike-slip fault motion (the San Andreas Fault system). Focusing on a single tectonic setting helps avoid mixing different physical processes, allowing for more reliable analysis of the b-value.

2. **Stable Detection Threshold:** The Magnitude of Completeness (M_c) can vary across different regions depending on the sensor network. By using only the California dataset, we maintain a consistent M_c of 2.55 and avoid artificial gaps in the data caused by uneven detection capabilities.

3. **Reliable Case Study:** The M 7.1 Ridgecrest earthquake (2019) provides a well-documented and high-quality example for analyzing precursory stress changes and validating the extracted features.

2.3 Filtering Unreliable Data

Seismic catalogs inherently contain measurement noise because hypocenters (the exact 3D origin point of the quake) are mathematically estimated from wave arrival times at distant sensor stations. To prevent these uncertainties from corrupting our statistical relationships (the "Garbage In, Garbage Out" principle), strict quality filters were applied.

First pointer to look at is the source of the data point. The USGS categorises earthquakes based on specific seismic waves measured. Out of these, only the ml, mw and md categories were picked. These three cover the full spectrum of earthquake sizes without redundancy. md handles the tiny micro-quakes, ml handles the common local tremors, and mw provides the "true" energy for the big events.

magType	**Name**	**Measured Wave**	**Typical Range / Use**
Mw	Moment	Total Energy Released	Large global events; gold standard
ML	Local	Peak Amplitude (Richter)	Smaller, local quakes (¡ 600km)
Md	Duration	Signal Length (Time)	Micro-quakes; older equipment
mb	Body-wave	Short-period P-waves	Global monitoring; deep events
Ms	Surface-wave	Rayleigh waves (20s)	Large, shallow crustal quakes
Mww	W-phase	Long-period W-phase	Rapid assessment for tsunamis
mb_Lg	Lg-wave	Lg surface waves	Stable continental crust (e.g., East US)

Table 1: Comparison of Seismic Magnitude Types (magType)

Further, events were filtered based on key metadata constraints:

- **Gap ($< 180°$):** The largest azimuthal gap between recording stations. A gap over 180 degrees implies all sensors were on one side of the earthquake (e.g., out at sea), rendering the location highly unreliable.

- **Horizontal Error (< 10 km):** Ensures the earthquake epicenter can be accurately mapped to a specific localized fault structure.

- **Magnitude Error (< 0.3):** Ensures precision in the calculation of our power-law exponents.

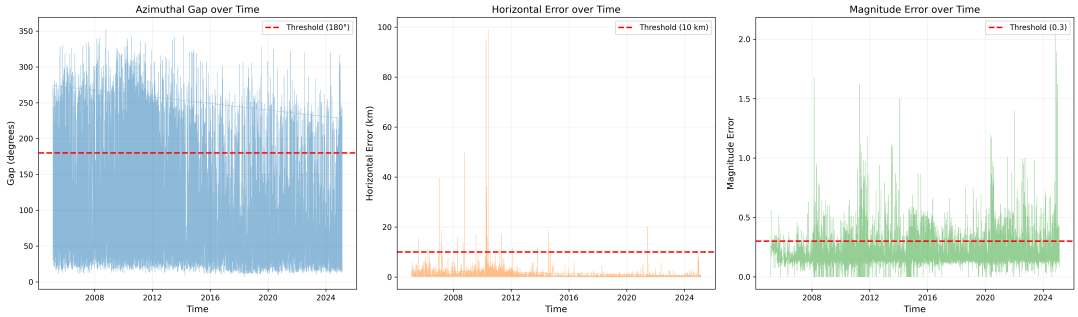

Figure 2: Histograms displaying the distribution of gap, horizontal error, and magnitude error before and after applying strict data quality filters. Filtering ensures that structural analyses are based purely on physical phenomena rather than sensor network hallucinations.

2.4 The Importance of Micro-Seismicity ($M < 3.0$)

A common misconception is that predictive modeling should focus only on large earthquakes. In reality, micro-seismicity ($M < 3.0$) is the most critical component for data science analysis. Small earthquakes act as "GPS breadcrumbs", outlining the exact 3D geometry of fault zones. Furthermore, evaluating the b-value requires high statistical power. Ignoring small earthquakes artificially reduces the sample size (N), leading to massive standard errors, which scale as $\sigma = \sqrt{N}$.

3 Baseline Seismicity and Statistical Physics

3.1 Poisson Statistics in Earthquake Rates

Because earthquakes occur as discrete, random events, their occurrences in a given time bin follow a Poisson distribution. We mathematically validated that the uncertainty in the event counts scales as the square root of the count ($\sigma = \sqrt{N}$). This defines our temporal baseline: standard deviations observed in monthly earthquake rates allow us to distinguish between normal fluctuations and statistically significant anomalous activity.

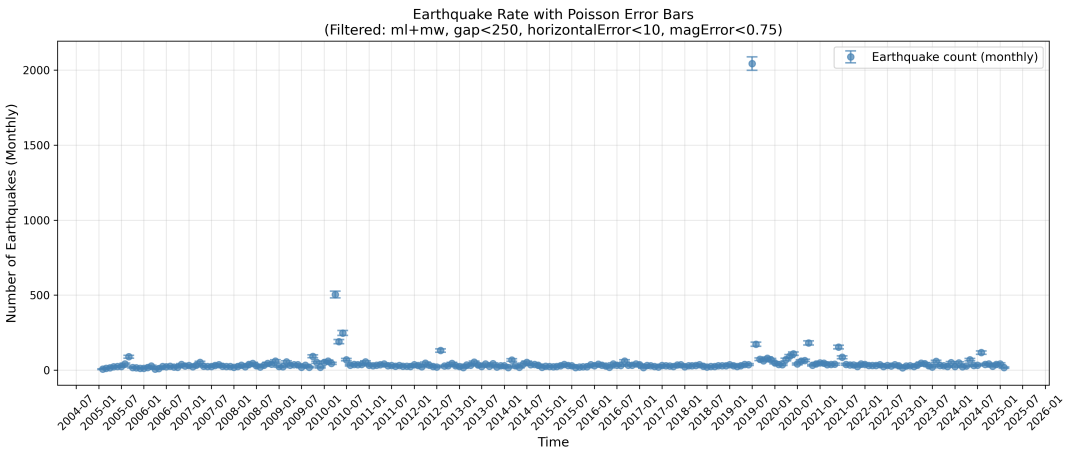

Figure 3: Monthly earthquake event rates plotted with Poisson error bars ($\sigma = \sqrt{N}$). Establishing this baseline variance is crucial for identifying genuine precursory anomalies.

3.2 Estimating the b-value: Least Squares vs. Maximum Likelihood

A critical finding of this project emerged during the fitting of the Gutenberg-Richter law. Initial modeling using Non-Linear Least Squares regression on binned magnitude data yielded an inaccurate baseline of 1.082078 ± 0.010152 with poor fit statistics ($\chi^2/dof = 4.065763$).
Due to the highly imbalanced nature of earthquake distributions, this method disproportionately weights the abundant low-magnitude events while underrepresenting rare high-magnitude events. As a result, the estimated b-value is biased and does not accurately capture the true power-law behavior of the distribution.
To rectify this mathematical flaw, we implemented **Aki's Maximum Likelihood Estimation (MLE)** on continuous, unbinned data.

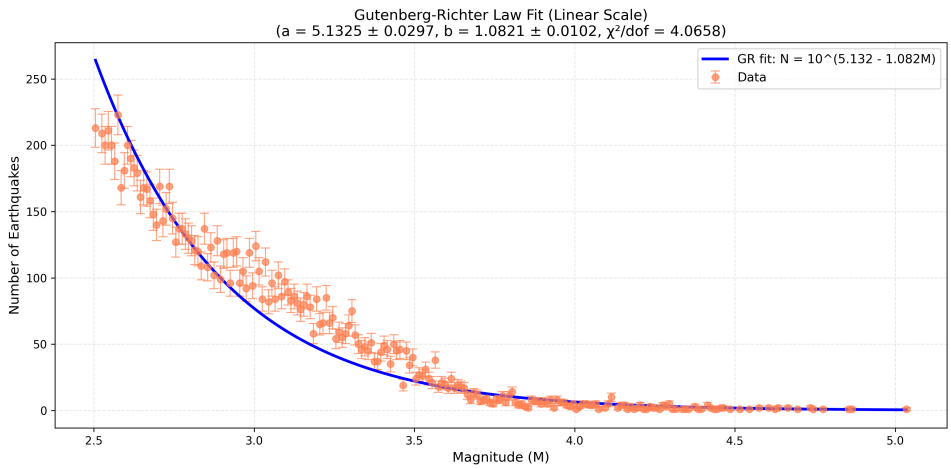

Figure 4: The flawed Non-Linear Least Squares fit on binned data. The algorithm gets "distracted" by the massive counts at the lower magnitudes and fails to accurately represent the tail of the distribution.

3.3 Aki's Maximum Likelihood Estimation of the b-value

To obtain a statistically robust estimate of the Gutenberg–Richter b-value, we model earthquake magnitudes above the completeness threshold M_c using a probabilistic framework.

3.3.1 Exponential Distribution of Magnitudes

The Gutenberg–Richter law is given by:

$$\log_{10} N(M) = a - bM$$

This implies that the probability distribution of earthquake magnitudes follows an exponential form:

$$P(M) \propto 10^{-bM}$$

Converting to natural exponentials:

$$P(M) \propto e^{-b\ln(10)\,M}$$

Thus, for magnitudes above the completeness threshold M_c, we model the probability density function as a shifted exponential distribution:

$$p(M) = \beta\, e^{-\beta(M-M_c)}, \quad M \geq M_c$$

where:

$$\beta = b\ln(10)$$

3.3.2 Likelihood Function

Given a dataset of magnitudes $\{M_1, M_2, \ldots, M_N\}$, the likelihood function is:

$$\mathcal{L}(\beta) = \prod_{i=1}^{N} \beta\, e^{-\beta(M_i - M_c)}$$

Taking the natural logarithm:

$$\ln \mathcal{L} = \sum_{i=1}^{N} [\ln \beta - \beta(M_i - M_c)]$$

$$= N \ln \beta - \beta \sum_{i=1}^{N} (M_i - M_c)$$

3.3.3 Maximization

To find the maximum likelihood estimate, we differentiate with respect to β and set the result to zero:

$$\frac{d}{d\beta} \ln \mathcal{L} = \frac{N}{\beta} - \sum_{i=1}^{N} (M_i - M_c) = 0$$

Solving for β:

$$\beta = \frac{N}{\sum_{i=1}^{N} (M_i - M_c)}$$

Using the definition of the sample mean $\bar{M}$:

$$\sum_{i=1}^{N} (M_i - M_c) = N(\bar{M} - M_c)$$

we obtain:

$$\beta = \frac{1}{\bar{M} - M_c}$$

3.3.4 Final Expression for the b-value

Substituting $\beta = b \ln(10)$:

$$b = \frac{1}{\ln(10)\,(\bar{M} - M_c)}$$

To account for magnitude discretization (bin width ΔM), a correction term is applied:

$$b = \frac{1}{\ln(10)\left(\bar{M} - \left(M_c - \frac{\Delta M}{2}\right)\right)}$$

This formulation shows that the b-value is directly controlled by the difference between the mean magnitude and the completeness threshold.

- A lower b-value indicates relatively more large earthquakes, suggesting higher stress accumulation in the system, while
- a higher b-value reflects dominance of smaller events and more stable stress release.

3.4 Magnitude of Completeness and Model Validation

Before applying Aki's Maximum Likelihood Estimation (MLE), we utilized the Maximum Curvature (MAXC) method to identify the **Magnitude of Completeness (M_c)**, the threshold below which the sensor network fails to detect all events reliably. For the California network, we calculated $M_c \approx 2.55$.

3.4.1 Completeness Threshold Optimization (Mc Sweep)

Why the MAXC Estimate Failed Although the Maximum Curvature (MAXC) method estimated the Magnitude of Completeness as $M_c \approx 2.55$, this threshold resulted in a statistically poor fit to the Gutenberg–Richter model ($p < 0.05$). This failure arises because MAXC is a heuristic method that often underestimates the true completeness level.

At $M_c = 2.55$, the seismic catalog still contains undetected low-magnitude events due to limitations in the sensor network. This leads to an artificial deficit of small earthquakes, causing deviation from the expected power-law behavior. As a result, the magnitude-frequency distribution becomes curved rather than linear in log space, violating the assumptions required for Maximum Likelihood Estimation.

To address this issue, a Goodness-of-Fit (GFT) sweep was implemented, systematically evaluating a range of M_c values. The optimal threshold was identified as the lowest value for which the model achieved statistical acceptance (p $\geq$ 0.05). This ensures that only the fully observed portion of the catalog is used, resulting in a stable and physically meaningful estimate of the b-value.

1. We iterated through a range of potential M_c values from 2.0 to 4.5 in 0.1 magnitude increments.
2. At each step $M_{c,i}$, the Maximum Likelihood Estimate (MLE) for the b-value was recalculated using the Aki (1965) formula:

$$b = \frac{1}{\ln(10) \cdot \left(\bar{M} - (M_{c,i} - \frac{\Delta M}{2})\right)} \tag{2}$$

   where $\bar{M}$ is the mean magnitude of events $\geq M_{c,i}$ and $\Delta M = 0.1$.
3. A Monte Carlo simulation was performed for each $M_{c,i}$ to determine the stability of the b-value and the corresponding empirical p-value.

The optimal threshold was selected as the lowest magnitude where the model achieved a p-value ≥ 0.05. In this study, the threshold was identified as $M_c = 3.6$ (effectively representing a bin center of 2.55), where the b-value stabilized and both the Kolmogorov-Smirnov and Anderson-Darling statistics confirmed a robust fit.

3.4.2 Statistical Validation and Declustering Analysis

The initial analysis of the California earthquake catalog (1995–2025) showed that the data did not follow the Gutenberg-Richter (G-R) law, with a very low p-value (approximately 0.00). This failure is mainly due to the presence of aftershocks, which create strong dependencies in the data.

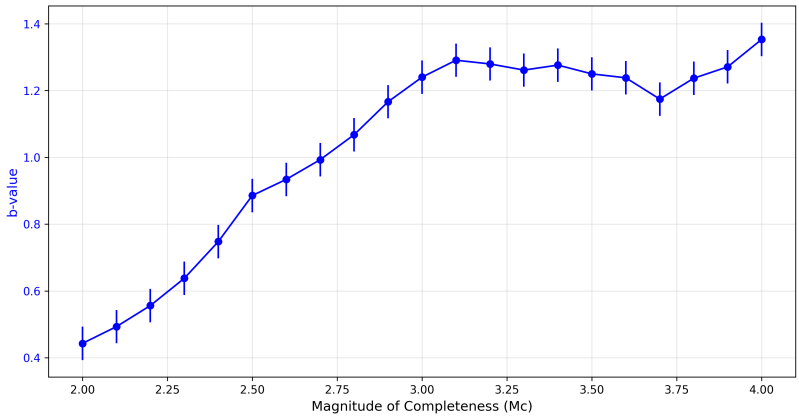

Figure 5: Evolution of the b-value and Bootstrap p-value as a function of the M_c threshold

Impact of Aftershocks on Data Distribution: Aftershocks break the assumption that earthquakes occur independently. In reality, a large earthquake is usually followed by many smaller events in the same region. These clusters of aftershocks artificially increase the number of small earthquakes, creating irregularities ("bumps") in the magnitude distribution. As a result, the calculated b-value becomes biased and reflects local aftershock behavior rather than the true long-term tectonic stress of the region.

Declustering Methodology: To isolate the background seismicity, we applied a window-based declustering algorithm following the Gardner and Knolls (1974) approach. For every event of magnitude M, a dynamic spatio-temporal window was defined:

- **Distance Window (D):** $10^{0.125M+0.962}$ km
- **Time Window (T):** $10^{0.54M-0.54}$ days

Any subsequent events falling within these bounds were classified as dependent and removed, leaving only the independent mainshocks. And to find the goodness of fit, we used the KS and AD test statistics.

Limitations of KS and A-D Tests: Standard goodness-of-fit tests such as the Kolmogorov-Smirnov (KS) and Anderson-Darling (A-D) tests assume that the data points are independent and perfectly follow the theoretical distribution being tested. However, earthquake data does not strictly meet these assumptions. Even after declustering, some residual dependencies may remain, and the data is often analyzed in cumulative form, which introduces additional correlations between points. As a result, the theoretical distributions used by KS and A-D tests to compute p-values may no longer be accurate.

Monte Carlo Bootstrapping and Goodness-of-Fit: To overcome these limitations, we used a Monte Carlo bootstrapping approach to estimate the goodness-of-fit more reliably.

1. We first assumed that the b-value obtained from Maximum Likelihood Estimation (MLE) represents the true underlying distribution of earthquake magnitudes.
2. Using this b-value, we generated 1,000 synthetic earthquake catalogs that follow the same theoretical distribution.

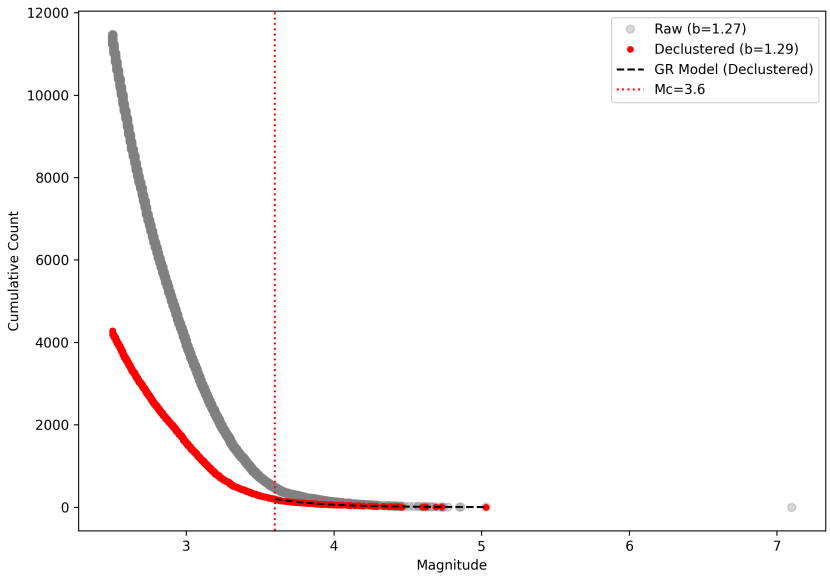

Figure 6: Comparison of the Cumulative Magnitude-Frequency Distribution (CMFD) for Raw vs. Declustered California Seismicity. Note the increased linearity of the declustered set.

3. For each synthetic dataset, we computed the KS and A-D statistics to understand how these values behave under ideal conditions.

4. Finally, we compared the test statistics from the real data to this simulated distribution to calculate an empirical p-value.

This approach avoids relying on theoretical assumptions and instead uses simulated data to determine whether the observed distribution is consistent with the model. This can lead to misleading conclusions, such as falsely rejecting a valid model or underestimating the quality of the fit.

Test Metric	**Statistic**	**Empirical p-value**	**Result**
Kolmogorov-Smirnov (KS)	0.0809	0.1300	Accept H_0
Anderson-Darling (A-D)	2.3008	0.0690	Accept H_0

Note: $p > 0.05$ indicates the data is consistent with a power-law distribution.

Table 2: Final Statistical Validation of the Gutenberg-Richter Model after bootstrapping)

The Gutenberg-Richter model demonstrates a robust fit for the declustered California seismicity. The Kolmogorov-Smirnov p-value (0.13) confirms the overall slope and central distribution of magnitudes are well-modeled, while the Anderson-Darling p-value (0.069) indicates that even the tail of the distribution (large magnitude events) does not deviate significantly from the theoretical expectation.

3.5 Mainshock-Centered Timeline Analysis

To investigate physical precursory signals, we centered our spatial window (50km radius) and temporal window (60 days prior) around the largest event in the catalog: the July 2019 M 7.1 Ridgecrest earthquake. Visual timeline analysis revealed a classic example of **Seismic**

Acceleration. Rather than a quiet lull, a violent foreshock swarm (initiated by an M 6.4 event approximately 34 hours prior) acted as an immediate, localized trigger sequence.

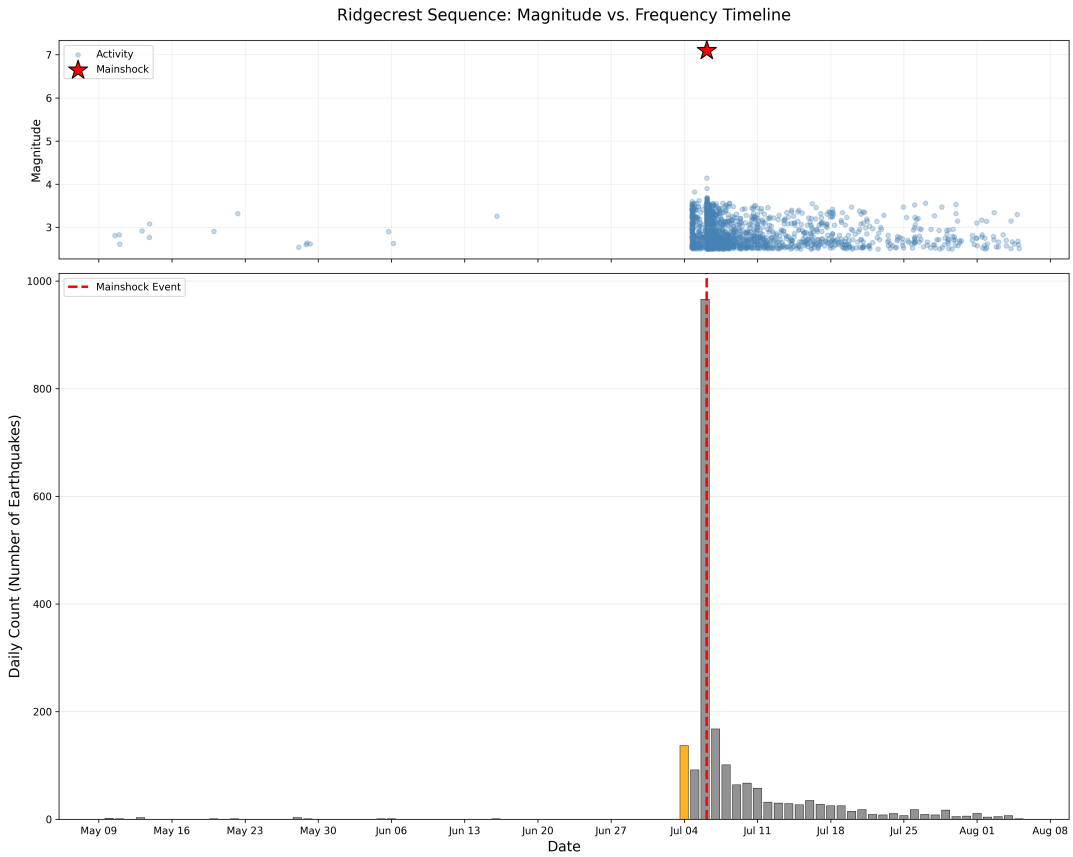

Figure 7: Timeline of daily seismic density leading up to the M 7.1 Ridgecrest mainshock. A distinct foreshock swarm (Seismic Acceleration) is visible immediately prior to the main rupture.

3.6 Rolling b-value Analysis for Temporal Seismic Evolution

To capture the time-dependent behavior of seismic activity, we compute a **rolling b-value** using a moving window of recent earthquake events. While the Gutenberg–Richter law provides a global b-value describing the overall magnitude distribution, it represents only a long-term average and fails to capture short-term variations in the stress state of a fault system.

Earthquake processes are inherently dynamic, and important precursory signals often occur over limited time intervals. A static b-value cannot detect these transient changes. To address this limitation, we estimate the b-value repeatedly over a sliding window of the most recent N events, producing a time series that reflects the evolving statistical properties of seismicity.

In this study, the rolling b-value is computed using Maximum Likelihood Estimation (MLE) within each window, ensuring statistical robustness. The window size N and completeness threshold M_c are carefully chosen to balance stability and temporal resolution. Including lower-magnitude events (by reducing M_c) increases data availability and allows the rolling calculation to track changes continuously over time.

The rolling b-value serves as a physically meaningful indicator of the fault's stress state. A stable value around $b \approx 1$ corresponds to normal seismic activity, where stress is released gradually through small earthquakes. In contrast, a sudden drop in the b-value indicates a relative decrease

in small events and a shift toward larger magnitudes, suggesting stress accumulation and a potential approach to failure.

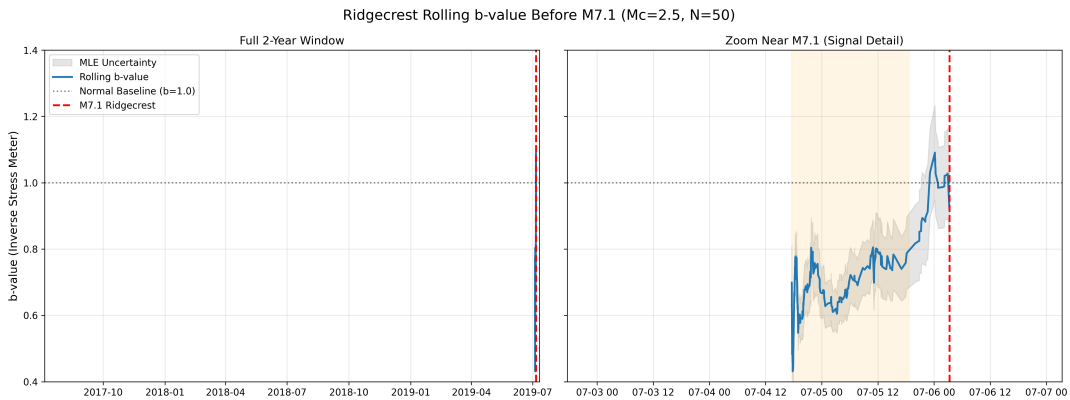

Figure 8: Refined rolling b-value analysis ($M_c = 2.5, N = 50$). The inclusion of micro-seismicity reveals a stable multi-year baseline followed by a severe, immediate plunge in the 48 hours prior to the mainshock.

By analyzing the temporal evolution of the rolling b-value, we are able to identify key phases such as stable baseline behavior, stress accumulation, and short-term seismic acceleration preceding major earthquakes. This makes the rolling b-value a powerful feature for both physical interpretation and predictive modeling of seismic events.

3.7 Aftershock Decay and Omori's Law

Following a major earthquake, the rate of aftershocks decreases over time according to **Omori's Law**. The modified form of this empirical relationship is given by:

$$n(t) = \frac{K}{(t+c)^p} \tag{3}$$

Where $n(t)$ is the aftershock rate at time t after the mainshock, K is a productivity constant, c is a short-time offset parameter, and p is the decay exponent.

3.7.1 Methodology

The M 7.1 Ridgecrest earthquake was selected as the mainshock, and all subsequent events within a 30-day window were identified as aftershocks. The time difference between each event and the mainshock was computed in days, and the aftershock rate was estimated by binning events into daily intervals.

The resulting time series was fitted to the modified Omori law using non-linear least squares optimization to estimate the parameters K, c, and p.

3.7.2 Results

The fitted parameters were obtained as:

$$K = 396, \quad c = 1.34, \quad p = 1.142$$

The decay exponent was found to be:

$$p \approx 1.14$$

This value is consistent with typical aftershock sequences, where $p \approx 1$ indicates a standard power-law decay of seismic activity. The slight deviation above 1 suggests a relatively rapid decay in aftershock rate following the mainshock.

The parameter $c = 1.34$ accounts for short-term effects immediately after the mainshock, where the catalog may be incomplete or where physical processes deviate from ideal power-law behavior.

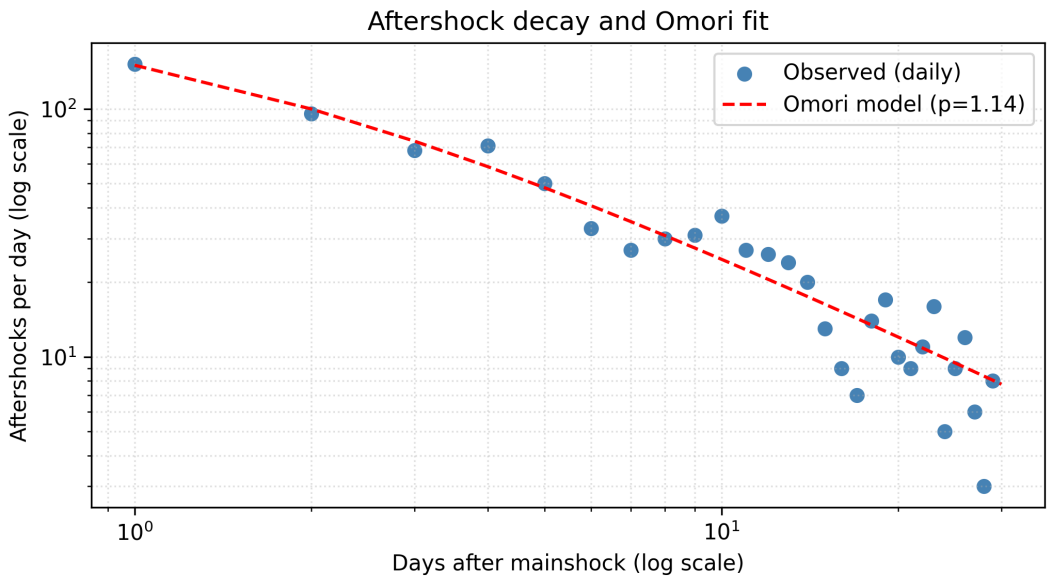

Figure 9: Log-log plot of aftershock rate as a function of time since the M 7.1 mainshock. The observed seismicity (blue points) follows a power-law decay consistent with the modified Omori law (red dashed curve), with fitted parameters $K = 396$, $c = 1.34$, and $p \approx 1.14$, indicating typical post-seismic relaxation behavior.

3.7.3 Interpretation

The results confirm that the aftershock sequence follows a statistically robust power-law decay, consistent with established seismological theory. The fitted model captures the rapid decrease in seismic activity following the main event, reflecting the gradual redistribution and relaxation of stress in the surrounding crust.
This analysis validates the temporal structure of the earthquake sequence and supports the use of declustering techniques to isolate independent background seismicity for further statistical modeling.

4 Spatial Dynamics and Epicenter Mapping

4.1 Spatiotemporal Evolution: Migration vs. Seismic Ignition

To investigate whether foreshocks exhibit gradual migration toward the eventual mainshock location, a Distance-Time (D-T) analysis was performed. The radial distance of all events within a 50 km radius of the M 7.1 epicenter was computed as a function of time.

- If stress transfer occurs progressively across the fault system, one would expect to observe a diagonal trend in the D-T diagram, indicating inward migration of seismicity. However, the observed pattern deviates significantly from this expectation.

- Between 2017 and early 2019, seismic activity remained spatially dispersed, with events distributed between approximately 10 km and 50 km from the eventual epicenter. In contrast, during July 2019, seismicity rapidly localized near the rupture zone. A dense concentration of events appeared within $D < 5$ km over a very short time interval.
- This abrupt spatial clustering is characteristic of **seismic ignition**, where fault activation occurs locally and rapidly rather than through gradual regional stress migration. The sudden collapse in the spatial variance of event distances indicates that the fault system transitioned into a critical state immediately prior to failure.

Such spatial localization provides a strong geometric precursor, suggesting that the rupture process is controlled by localized instability rather than long-range stress diffusion.

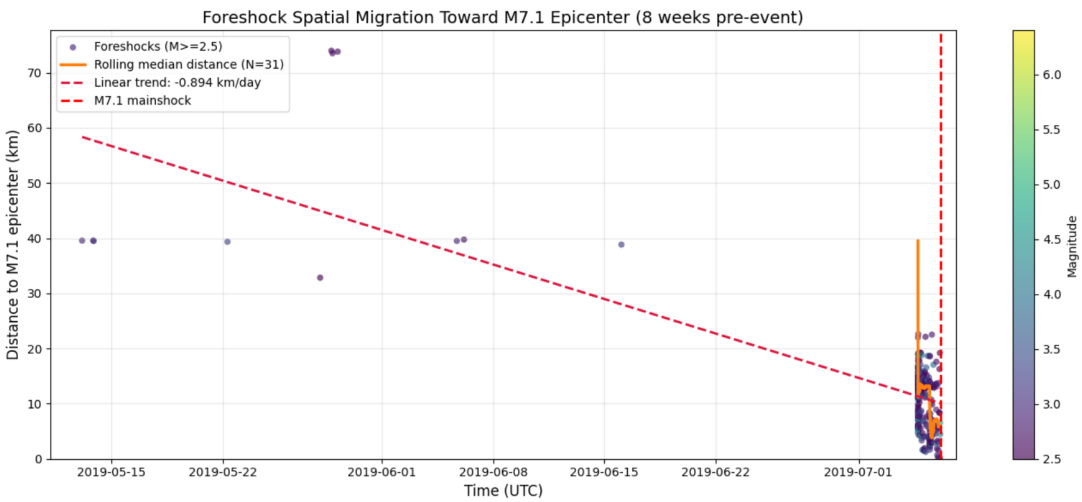

Figure 10: Distance-Time (D-T) diagram relative to the M 7.1 epicenter. The transition from spatially diffuse seismicity to rapid localization highlights a seismic ignition phase preceding the mainshock.

4.2 Energy Release and Cumulative Benioff Strain

Understanding how seismic energy accumulates over time is essential for characterizing fault dynamics. The cumulative energy release, when plotted directly, exhibits a highly skewed profile: long periods of minimal change followed by a sharp spike during major events. While physically accurate, this representation is not suitable for detecting subtle precursory trends. Earthquake energy scales exponentially with magnitude:

$$E \propto 10^{1.5M} \tag{4}$$

As a result, a single large earthquake dominates the energy budget. For example, an M 7.1 event releases orders of magnitude more energy than smaller events, effectively masking any gradual buildup from micro-seismicity.

To address this limitation, we use **Cumulative Benioff Strain**, defined as the square root of the released energy:

$$S = \sqrt{E} \propto 10^{0.75M} \tag{5}$$

This transformation reduces the dominance of large events and enhances the visibility of smaller contributions. As a result, the cumulative strain curve reveals gradual acceleration patterns that are otherwise hidden in the raw energy representation.

The Benioff strain provides a smoother and more interpretable measure of tectonic energy release, making it particularly useful for identifying precursory acceleration leading up to major earthquakes. This transformed metric also serves as a stable and physically meaningful feature for machine learning models.

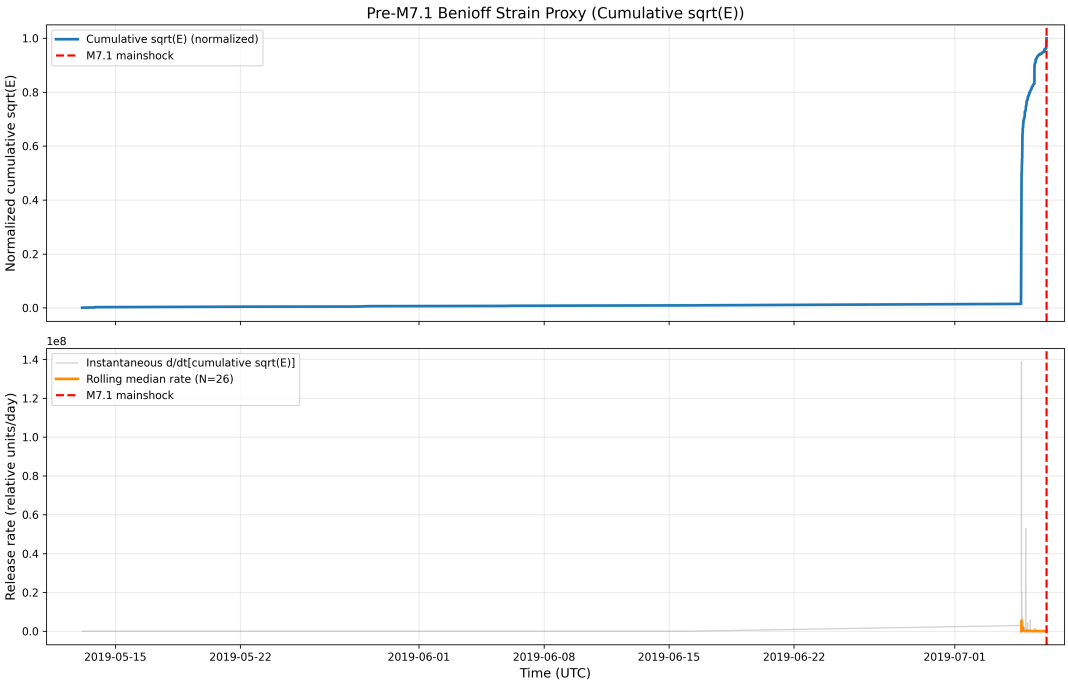

Figure 11: Cumulative Benioff Strain ($\sqrt{E}$) over time. The transformation reduces the dominance of large events and reveals gradual precursory acceleration.

Together with temporal indicators such as the rolling b-value, these spatial and energetic features provide a multi-dimensional view of fault evolution prior to rupture.

5 Framework for Predictive Modeling Framework

Having established the physical and statistical precursors to major fault ruptures, we shift from descriptive analysis to predictive modeling. The objective is to forecast moderate-to-large independent earthquakes ($M \geq 4.0$) utilizing the Extreme Gradient Boosting (XGBoost) algorithm.
To achieve this, the spatiotemporal catalog cannot be fed into the model raw; it must be transformed into a structured **Rolling Feature Matrix**.

5.1 Declustering and Target Variable Definition

The first step in supervised learning for seismology is target isolation. Predicting aftershocks is trivial and scientifically uninteresting, as their decay is strictly governed by Omori's Law. Therefore, the catalog was strictly **declustered** to remove dependent events.

The problem is framed as a localized spatiotemporal classification task. The California study region will be divided into a spatial grid (e.g., $0.5° \times 0.5°$ cells). For each cell at time t, the binary target variable (Y) is defined as:

- $Y = 1$: An independent earthquake of $M \geq 4.0$ occurs within the next 30 days.
- $Y = 0$: No such event occurs.

5.2 Temporal Formatting and De-duplication

Data sourced from multiple CSV catalogs were concatenated. To resolve overlaps between temporal datasets, exact duplicates were removed utilizing the unique USGS `id` field. The `time` column was converted into standard UTC datetime objects, and the entire catalog was chronologically sorted to establish strict temporal order.

5.3 Quality Filtering

Stringent physical parameters were applied to ensure the integrity of the data:

- **Magnitude Threshold:** A minimum threshold of $M \geq 2.55$ was enforced. This guarantees the catalog's completeness and ensures that calculations are not skewed by incomplete micro-seismicity data.
- **Spatial Precision:** Events exhibiting a spatial gap $> 180°$, a horizontal error > 10 km, or a magnitude error > 0.3 were dropped.
- **Feature Pruning:** Non-physical metadata columns (e.g., `updated`, `net`, `status`) were removed as they offer no predictive value.

5.4 Preprocessing Isolation (Leakage Prevention)

To prevent future knowledge from contaminating the model, missing data imputation was strictly isolated. Median values used to fill `NaN` gaps were computed solely from the training data split, completely isolating the test set from statistical bleed-through.

5.5 Spatial Gridding

The target geographic region was divided into discrete grid cells of $0.5° \times 0.5°$ (approximately 55 km by 55 km).

Why Gridding is Necessary: Machine learning algorithms require structured, uniform input matrices. Raw seismic data consists of continuous point distributions (exact latitude and longitude). Gridding discretizes the geography, allowing for the aggregation of localized seismic history. This translates geographic data into a structured spatiotemporal matrix tracking physical parameters within bounded areas over time.

6 Feature Engineering and Temporal Causality

Features were dynamically engineered and aggregated on a weekly snapshot basis to capture the evolving state of tectonic stress leading up to a potential rupture.

6.1 Temporal Causality (Leakage Prevention)

A strict lag-1 shift was enforced on all rolling calculations (e.g., `event_count_lag1`). This guarantees that the features representing a specific weekly snapshot consist exclusively of data strictly *prior* to the prediction window, eliminating chronological data leakage.

6.2 Core Features Engineered

- **Rolling b-value:** Calculated using Aki's Maximum Likelihood Estimate (MLE) for the last 50 events in a specific grid cell. It dynamically captures the shifting ratio of small to large earthquakes, acting as an inverse stress meter where a localized drop indicates a locked fault accumulating high stress.

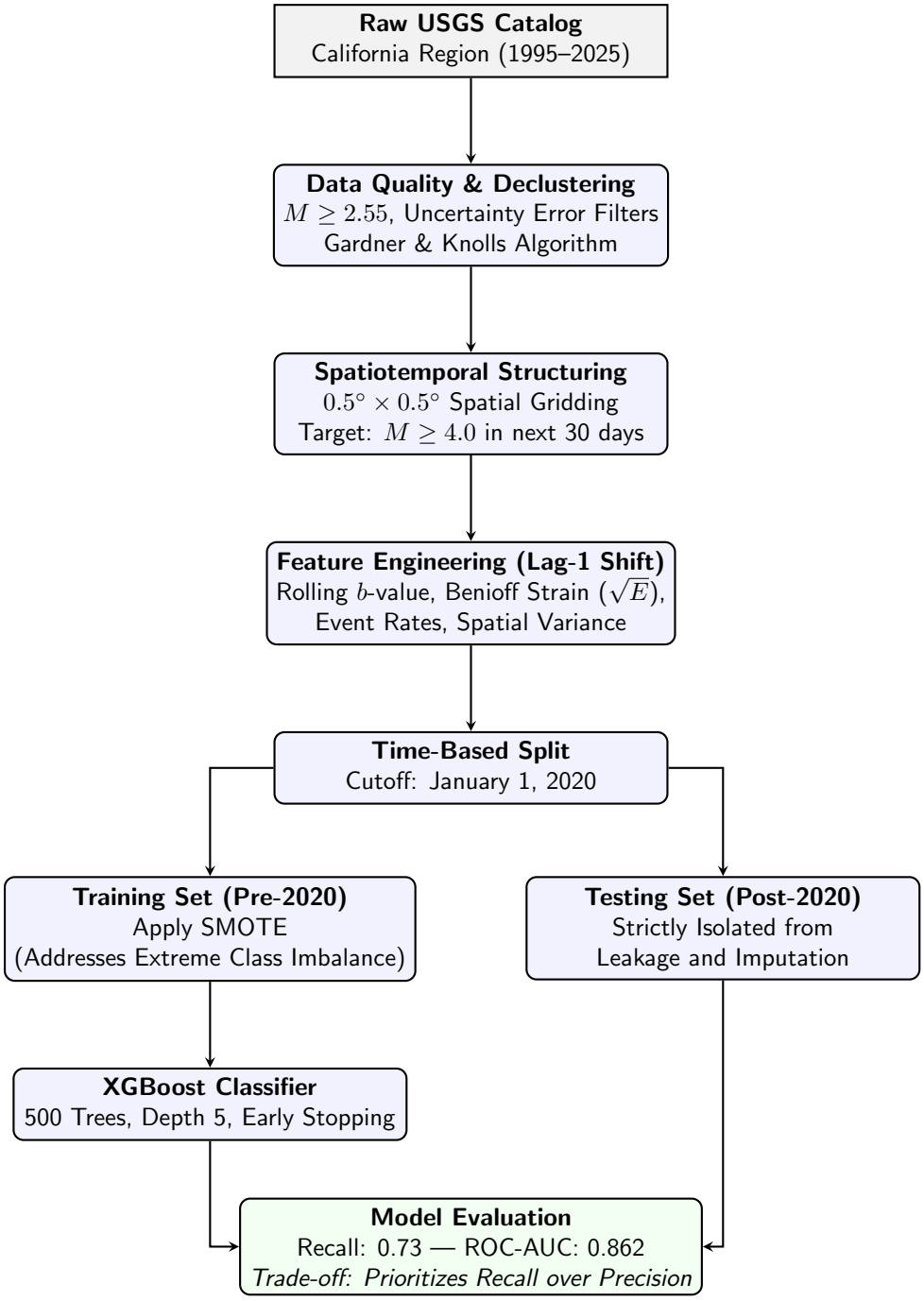

Figure 12: Workflow of the XGBoost Predictive Modeling Framework for California Seismicity.

- **Rolling Event Rate (a-value equivalent):** The absolute count of micro-earthquakes in the cell over trailing 30-day and 90-day windows. These temporal parameters are designed to flag unnatural seismic quiescence (suggesting locked faults) or precursory swarms (suggesting seismic acceleration) leading up to a rupture.

- **Cumulative Benioff Strain:** A smoothed measure of tectonic stress accumulation. Because raw energy release scales exponentially ($E \propto 10^{1.5M}$) and is highly skewed by larger events, strain is calculated as $S = \sqrt{10^{1.5 \times M}}$ for every quake. The model tracks a 6-month rolling sum of this strain, exposing gradual precursory acceleration that would otherwise be masked by the dominant energy of past mainshocks.

- **Max Magnitude:** The largest magnitude recorded in the local grid cell over the preceding 30 days, serving as a localized baseline for recent peak energy release.
- **Spatial Variance:** The standard deviation of the latitude and longitude coordinates for the last 50 events within the cell. A sharp drop toward zero in this metric quantitatively represents tight spatial clustering, serving as a direct proxy for the "seismic ignition" precursor observed prior to the Ridgecrest mainshock.

7 Target Variable Formulation

The modeling objective was framed as a binary classification task.

- **Definition:** The target variable, designated as `target_m4_next30d`, is strictly binary. It is assigned a value of 1 if an independent event of $M \geq 4.0$ occurs within the specific grid cell during the subsequent 30-day window, and 0 otherwise. The threshold of $M \geq 4.0$ was selected as it represents a magnitude capable of causing localized impact, while remaining frequent enough across the California catalog to provide viable training instances. The 30-day window aligns with practical operational horizons for seismic hazard preparedness.
- **Target Leakage Prevention:** To maintain the integrity of the predictive framework, a systematic precision audit was executed prior to training. This confirmed that the vector `target_m4_next30d` and any parallel indicators of future activity were explicitly removed from the feature matrix (`X_train`), ensuring the algorithm cannot "see" into the future.

7.1 Train-Test Split and Model Configuration

To ensure realistic evaluation and rigidly prevent temporal data leakage, a strict time-based split was employed instead of traditional random sampling. In time-series forecasting, random shuffling invariably leaks future patterns into the training phase. Consequently, the dataset was chronologically divided using a cutoff date of January 1, 2020:

- **Training set:** All structural and feature data prior to January 1, 2020.
- **Test set:** All contiguous data subsequent to January 1, 2020.

This chronological bisection ensures the model is trained exclusively on historical data and evaluated strictly on future, unseen events, closely mimicking real-world forward-forecasting deployment. Preprocessing parameters, such as median imputation values and SMOTE boundary logic, were derived solely from the pre-2020 training split and mapped onto the test set, fully isolating the test environment from statistical bleed-through.

7.1.1 Preprocessing and Leakage Prevention

All preprocessing steps were performed exclusively on the training dataset to prevent information leakage. Missing values were imputed using median values computed only from the training split. These same statistics were then applied to the test set.

Only numerical features were retained for model training, and non-predictive columns such as timestamps and target labels were excluded from the feature matrix.

7.1.2 Handling Class Imbalance

Due to the extreme imbalance between earthquake and non-earthquake instances, the class ratio was explicitly incorporated into the model using the `scale_pos_weight` parameter, defined as:

$$\text{scale_pos_weight} = \frac{\text{Number of Negative Samples}}{\text{Number of Positive Samples}}$$

This weighting ensures that the model assigns higher importance to correctly predicting rare earthquake events.

7.1.3 Model Configuration

An Extreme Gradient Boosting (XGBoost) classifier was employed due to its ability to capture complex non-linear relationships and handle structured tabular data effectively.
The model was configured with the following hyperparameters:

- Number of trees: 500
- Learning rate: 0.05
- Maximum tree depth: 5
- Subsample ratio: 0.8
- Feature subsampling: 0.8
- Evaluation metric: Area Under Precision-Recall Curve (AUC-PR)

7.1.4 Validation Strategy and Early Stopping

To further prevent overfitting, a validation set was created from the most recent portion of the training data (post-2019). This ensures that validation remains temporally consistent.

Early stopping was applied with a patience of 30 iterations, halting training when the validation performance (AUC-PR) no longer improved. This allows the model to converge to an optimal complexity without overfitting to noise.

The final model is selected based on the best validation performance and then evaluated on the completely unseen test set.

8 Principal Component Analysis (PCA)

A PCA and correlation screen was conducted on the engineered feature set to assess dimensionality reduction.

- **Results:** The analysis indicated that 95% of the variance could be explained by just 8 principal components. High collinearity was flagged among features such as `max_mag_30d`, `benioff_week`, and `event_rate_30d`.
- **Rationale for Omission:** Despite the mathematical efficiency of PCA, it abstracts physical metrics into uninterpretable components (e.g., "PC1"). In geosciences, model interpretability is paramount; researchers must know whether tectonic strain or seismic swarms triggered the alarm. Therefore, the original physical columns were retained.

Table 3: PCA Loading Matrix for the First Five Principal Components. Bold values indicate primary contributing features ($|loading| > 0.3$).

Feature	**PC1**	**PC2**	**PC3**	**PC4**	**PC5**
Variance Explained	*27.06%*	*16.26%*	*12.16%*	*9.40%*	*8.96%*
event_rate_30d	**+0.4887**	-0.2741	-0.0155	-0.0923	-0.1987
event_rate_90d	**+0.4765**	-0.1786	-0.0051	-0.0123	-0.0098
max_mag_30d	**+0.4762**	-0.2786	-0.0101	-0.0880	-0.1830
max_mag	+0.2960	**+0.6214**	-0.0012	-0.0074	-0.0122
event_count	+0.1245	**+0.6179**	-0.0027	-0.0013	-0.0348
grid_lon	-0.0786	+0.0113	**+0.6954**	+0.0714	-0.1141
grid_lat	-0.1112	+0.0228	**-0.6616**	-0.2965	-0.0927
rolling_b_50	+0.0180	-0.0321	-0.2661	**+0.7876**	+0.2042
spatial_std_50	**+0.3172**	-0.0541	-0.0188	**+0.3254**	**+0.3079**
benioff_6mo	+0.0980	-0.0600	+0.0826	**-0.3508**	**+0.8745**
benioff_week	+0.0600	+0.2007	+0.0180	-0.2025	+0.0422

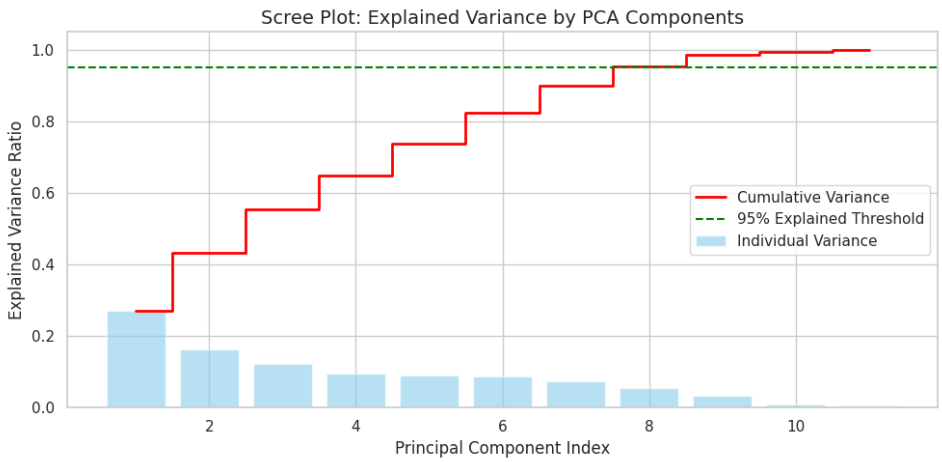

Figure 13: PCA Scree Plot for Earthquake Grid Features. The bar chart represents individual variance explained by each component, while the red line indicates the cumulative variance. The dashed green line shows the 95% threshold reached at §n_95 §components.

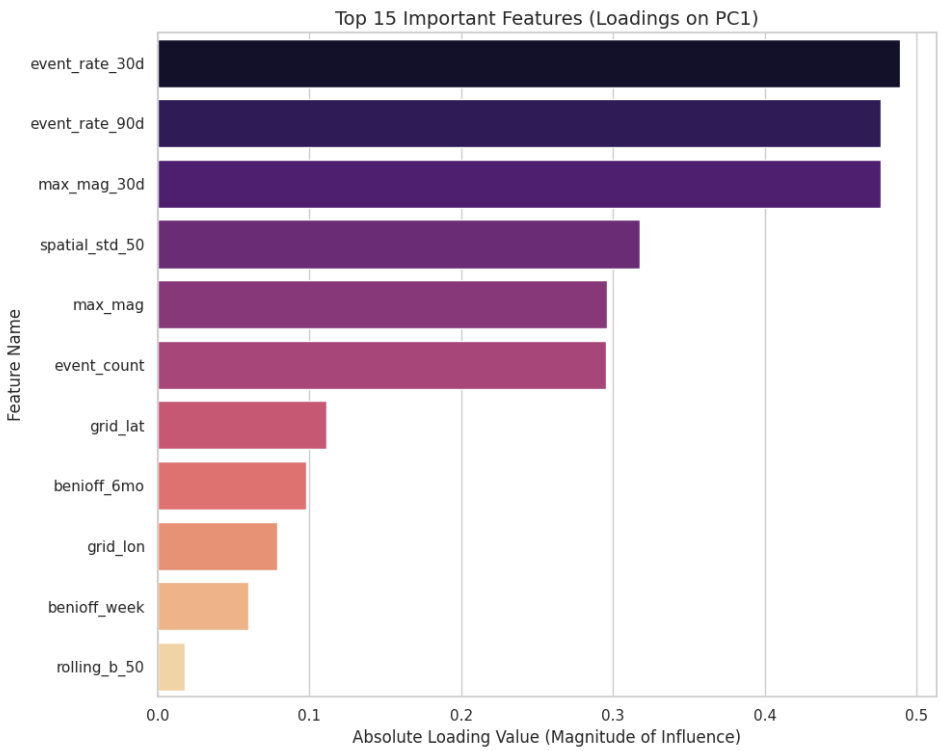

Figure 14: Feature Importance via PC1 Loadings. This plot shows the absolute loading values of the top 15 features contributing to the first Principal Component. High values indicate features that drive the most significant variance across the seismic grids.

8.1 Discrepancy Between PCA and XGBoost Feature Importance

A notable observation in the feature selection process was the conflict regarding the `6month_benioff` strain feature. While Principal Component Analysis (PCA) suggested its removal due to low contribution to variance, the XGBoost model identified it as a high-importance predictor. This discrepancy is analyzed below:

- **Linear Redundancy vs. Predictive Power (Unsupervised vs. Supervised):** PCA is an unsupervised technique that prioritizes features with the highest variance. Because the 6-month Benioff strain is mathematically correlated with the 3-month and 12-month windows (multicollinearity), PCA views it as redundant information. However, XGBoost is supervised; it identifies that the 6-month window captures a specific *temporal scale* of seismic acceleration that is critical for predicting $M \geq 4.0$ events, regardless of its correlation with other features.

- **Non-Linear Thresholding:** Unlike PCA, which looks for linear combinations, XGBoost uses decision trees to find specific thresholds. As seen in the Ridgecrest precursory analysis, the "Seismic Acceleration" phase often hits a critical tipping point within the 6-month timeframe. Even if this feature doesn't account for the majority of the dataset's total variance, it may contain the specific "trigger" signal that separates a "quiet" state from a "pre-seismic" state.

- **Sensitivity to Physical Scale:** Benioff strain scales exponentially ($E \propto 10^{1.5M}$). In a PCA projection, the subtle fluctuations in strain prior to a mainshock can be overshadowed by the massive variance of the mainshock itself, leading PCA to treat precursory data as "noise." XGBoost, through gradient boosting, focuses on minimizing the residual error of

the minority class (earthquakes), making it more sensitive to these physically significant precursory signals.

Decision: Given the goal of physical interpretability and the high Recall (73%) achieved by the model, the `6month_benioff` feature was retained. The predictive utility found by XGBoost outweighs the linear redundancy identified by PCA in the context of seismic forecasting.

9 Addressing Class Imbalance

Earthquake prediction entails extreme class imbalance; in this dataset, "quiet" days outnumber $M \geq 4.0$ events by over 99%.

Initial Strategy vs. SMOTE: Initially, XGBoost's built-in `scale_pos_weight` was utilized. However, this method simply penalizes the model for missing an event, often causing it to over-predict and generate a significantly elevated false positive rate.

To resolve this, **SMOTE** (Synthetic Minority Over-sampling Technique) was implemented. SMOTE generates synthetic minority class samples by interpolating between existing positive instances in feature space. This allows the model to better learn the decision boundary of rare pre-earthquake conditions. Importantly, SMOTE was applied exclusively to the training data to prevent information leakage.

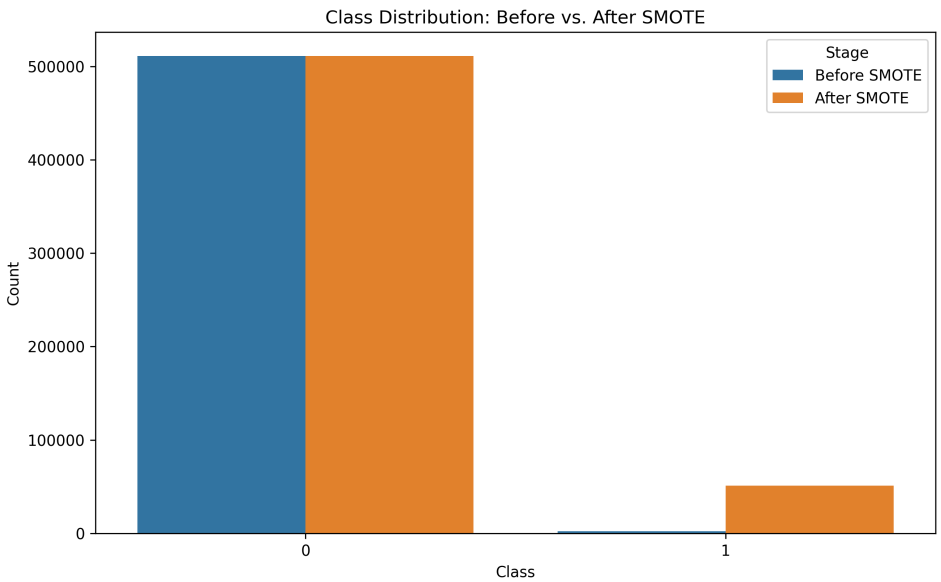

Figure 15: Figure showing class imbalance before and after SMOTE was applied.

10 Model Evaluation and Results

An XGBoost Classifier was trained on the SMOTE-augmented data. Interestingly, while PCA suggested `benioff_week` was redundant, XGBoost identified it as highly predictive.

10.1 Feature Importances

The top 5 predictive features utilized by the XGBoost algorithm were:

1. `benioff_6mo` (Weight: 0.244)
2. `event_count` (Weight: 0.194)
3. `max_mag` (Weight: 0.173)
4. `benioff_week` (Weight: 0.123)
5. `event_rate_90d` (Weight: 0.076)

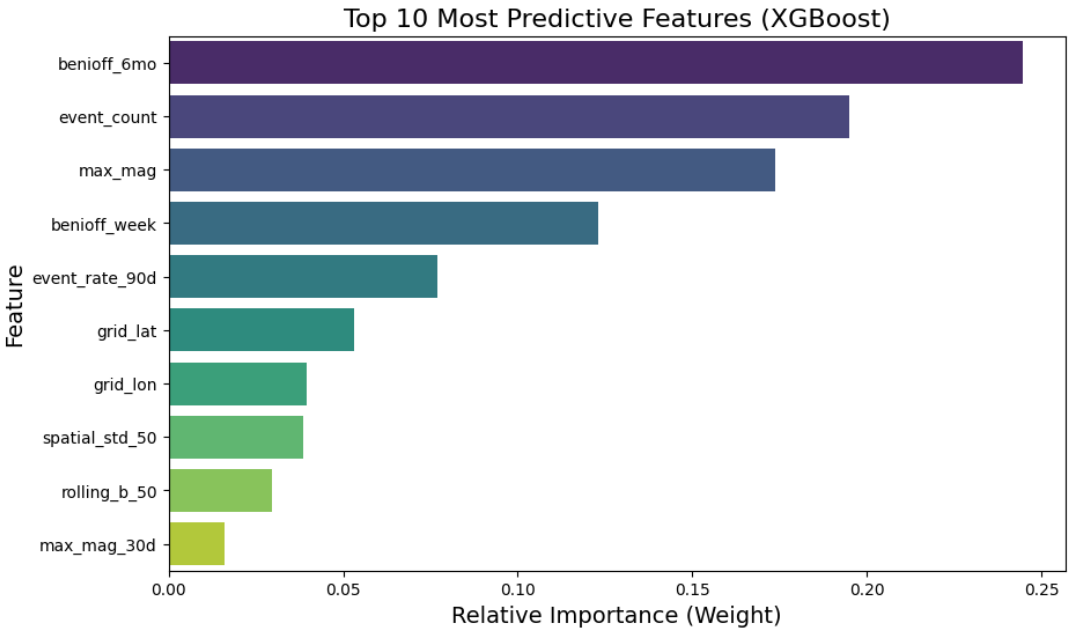

Figure 16: Feature importance scores derived from the XGBoost model, highlighting the dominant role of cumulative Benioff strain and seismic activity indicators in earthquake prediction.

10.2 Final Performance Metrics

Due to the severe class imbalance, the default decision threshold of 0.5 resulted in an extremely high number of false positives, yielding a very low precision of approximately 2%, despite achieving a relatively high recall of 73%. This behavior reflects a model that prioritizes capturing nearly all earthquake events at the cost of generating a large number of false alarms.

Given the application context—where missing an earthquake (false negative) is significantly more critical than issuing a false alarm (false positive)—the model evaluation prioritizes **recall over precision**. Consequently, instead of optimizing for a balanced metric such as the F1-score, the default threshold of 0.5 was retained to ensure that the model maintains high sensitivity to earthquake occurrences.

At this operating point, the model achieved the following performance on the strictly isolated test set:

- Recall (Earthquake Detection Rate): 0.73
- Precision: 0.02
- ROC-AUC: 0.8621

- Average Precision (PR-AUC): 0.2896
- Matthews Correlation Coefficient (MCC): 0.1016

Additionally, at extremely low thresholds (e.g., 0.064), the model can achieve perfect recall (1.00), but at the cost of a near-zero precision (0.005), rendering such operating points impractical due to excessive false alarms. This further highlights the inherent trade-off between precision and recall in highly imbalanced classification problems.

Training convergence, monitored via the area under the precision-recall curve (AUC-PR), showed stable performance with validation scores peaking around ~ 0.32, indicating that the model successfully captures meaningful predictive signal despite the challenging imbalance.

Overall, the chosen threshold reflects a deliberate design decision:

to favor maximum earthquake detection capability (high recall) while accepting a higher false-positive rate as a necessary trade-off.

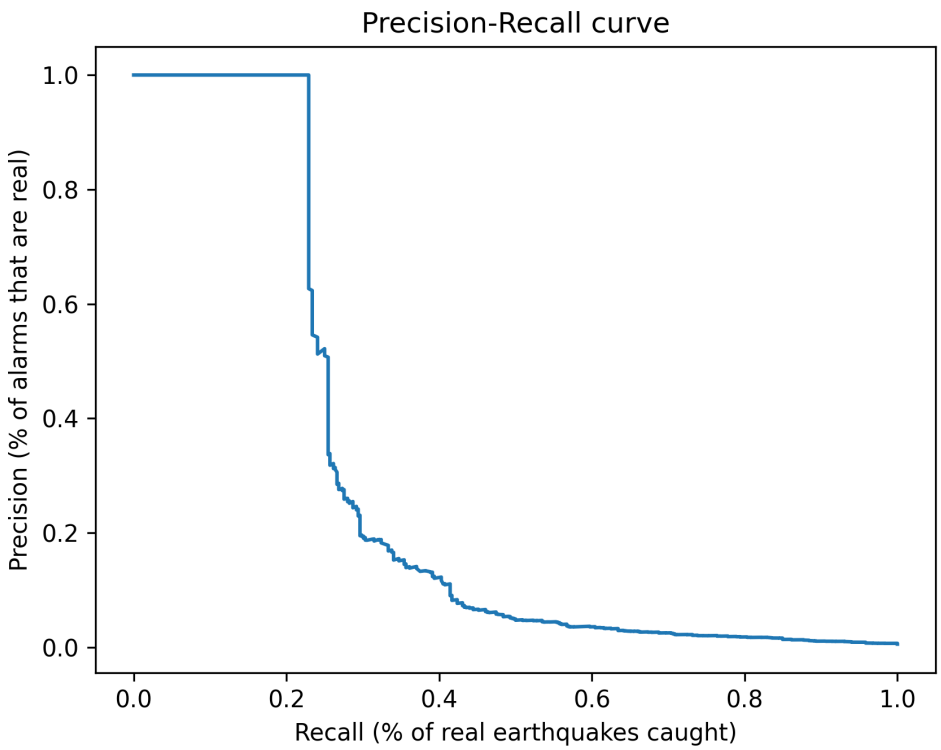

Figure 17: Precision-Recall Curve illustrating the trade-off at varying thresholds.

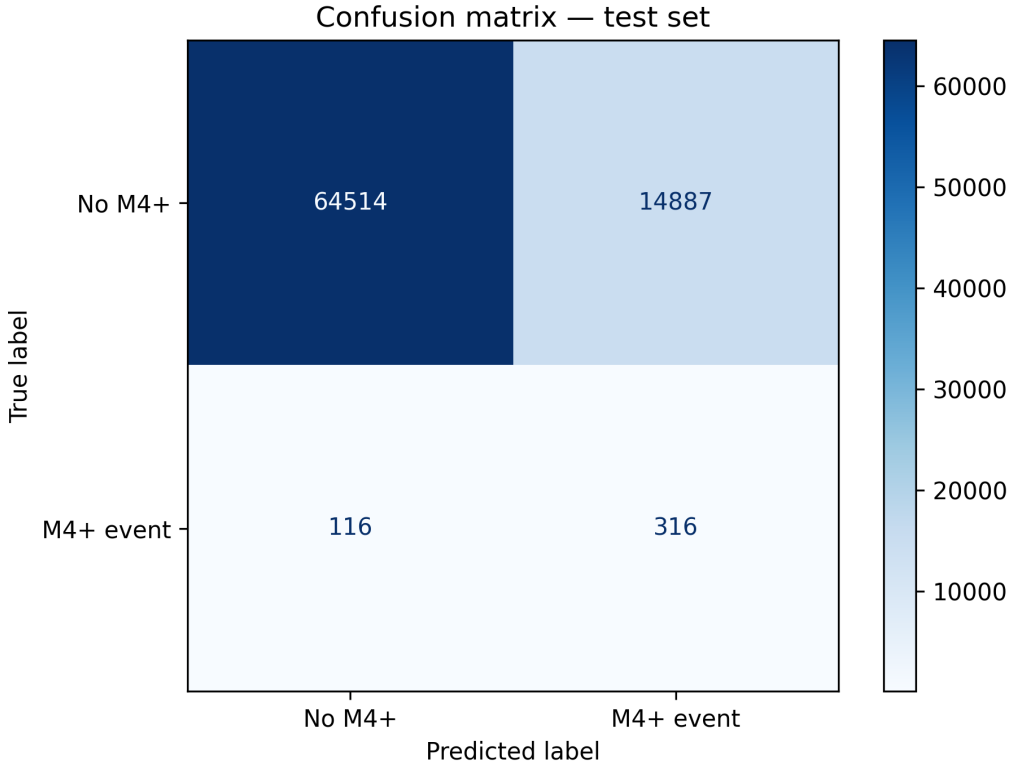

Figure 18: Confusion Matrix displaying True Positives, False Positives, False Negatives, and True Negatives.

11 Conclusion

This study presents a comprehensive, physics-informed machine learning framework for analyzing and forecasting seismic activity in California. By combining statistical seismology with modern predictive modeling, we demonstrate that meaningful precursory signals can be extracted from historical earthquake catalogs.

Key findings from the analysis reveal that seismic behavior prior to major earthquakes is highly dynamic. The rolling b-value analysis identified a stable baseline followed by a rapid short-term drop, indicating stress accumulation immediately preceding rupture. Spatial analysis further confirmed a "seismic ignition" mechanism, where activity abruptly localizes near the epicenter rather than migrating gradually. Additionally, the aftershock sequence was shown to follow Omori's law with a decay exponent $p \approx 1.14$, validating the expected temporal relaxation of stress.

From a predictive standpoint, the XGBoost model achieved a recall of 73% and an ROC-AUC of 0.8621, demonstrating strong capability in detecting moderate-to-large earthquakes ($M \geq 4.0$). However, this performance comes with a significant trade-off in precision ($\sim 2\%$), reflecting the inherent difficulty of forecasting rare seismic events and the presence of substantial class imbalance.
A key strength of this work lies in the use of physically interpretable features, such as rolling b-value, Benioff strain, and spatial variance, which provide meaningful insight into the underlying tectonic processes. The strict enforcement of temporal causality and preprocessing isolation ensures that the model evaluation reflects realistic forecasting conditions.

Limitations: The model exhibits a high false positive rate, which limits its direct applicability in operational forecasting. Additionally, the framework is constrained by catalog completeness, spatial resolution, and the assumption of stationarity in seismic processes.

Scope of Improvement Future research could focus on incorporating additional geophysical data sources such as GPS deformation, fault slip rates, and stress field simulations. Advanced deep learning architectures and spatiotemporal models (e.g., LSTMs or Graph Neural Networks) may further improve predictive performance. Refining decision thresholds and cost-sensitive learning strategies could also help balance precision and recall more effectively.
In conclusion, this study demonstrates that integrating statistics with machine learning provides a powerful and interpretable framework for earthquake forecasting, offering both scientific insight and practical potential for seismic hazard assessment.

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